Sean Lin

909-538-7519 seanlin@stanford.edu LinkedIn Portfolio

Education

Stanford University **Sep. 2023 – Dec. 2026**
M.S. in Aeronautical and Astronautical Engineering (3.86/4.00) *Stanford, CA*

- Relevant Coursework: Classical Dynamics, Orbital Mechanics, Advanced Feedback Control Design, Optimal Control
- Note: Took a Leave-of-Absence from Jan 2024 to April 2025

University of California, Berkeley **Aug. 2019 – May 2023**
B.S. in Mechanical Engineering, B.A. in Business Administration, Minor in EECS (3.93/4.00) *Berkeley, CA*

- M.E.T. Program; High Honors; Regents and Chancellor's Scholar; ASME; Tau Beta Pi

Experience

HyLight **September 2024 – March 2025**
Airship Dynamics Engineering Intern *Paris, FR*

- Designed a bench that reads thrust, torque, vibration, RPM, and power data from novel electric-propulsion-systems.
- Characterized aerodynamic coefficients of airship across attitudes and airspeeds for flight controller implementation.
- Reverse-engineered a ballonet system for buoyancy control. Sensor input from altimeter and temperature sensor.

De-Ice **June 2024 – Aug 2024**
Thermals and Dynamics Intern *Somerville, MA*

- Cradle-to-the-grave design and assembly of a bench test setup to mimic in-flight avionics thermal conditions. Applied first principles of fluids and heat transfer for analytical validation.
- Perform vibration analysis on avionics. Propose solutions for 1st or 2nd order resonant coupling problems.

NASA Ames Research Center **September 2024 – May 2025**
Mechanical Engineer (Part Time) *Moutain View, CA*

- Digitized 100+ mechanical drawings for the Electric Arc Shock Tube Facility (EAST).
- Perform structural analysis on high-pressure parts prior to design/material changes.

SpaceX **June 2023 – September 2023**
Associate Avionics Engineer *Hawthorne, CA*

- Design a Destructive Physical Analysis routine on avionics EEE parts. Findings lead to a containment of $45000.
- Perform electrical Failure Analysis (FA) of pressure transducers and capacitors.
- Validated Raptor engine MEMS and piezoelectric oscillators under helium-rich environments using Physics-of-Failure.

SpaceX **May 2022 – August 2022**
Launch Engineering Intern *Cape Canaveral, FL*

- Full design ownership of 4 Ground-Station-Equipment and launch pad hardware assemblies.
- Closed any open supply chain technical issues for all sleeve bearings used on the launch pad.

Research

Institute for Flight Mechanics and Control **Jun 2026 – Present**
Visiting Researcher - PI: Cunis Torbjorn *Stuttgart, Germany*

- Controllability-based design of VLEO satellites in rarified atmospheric orbit.
- Designed an optimization workflow for connecting control-surface configuration with desired vehicle torque.
- Defined key torque gradients to implement in the controller.

Space Rendez-Vous Lab **Jan 2025 – Present**
Student Researcher - PI: Simone D'Amico *Stanford, CA*

- Design a vision-based satellite navigation workflow for improved space-situational-awareness (SSA).
- Propagate orbits of a catalog of 20,000+ satellites and perform a detectability analysis on visible satellites.
- Designed a catalog-refining feedback workflow where observer and target satellites reinforce each-others positions.

Skills

Software and Embedded Systems: Python, MATLAB, C++, Java, Julia, SQL, ESP32, Arduino, Raspberry Pi
Engineering: SolidWorks, ANSYS, NX, AutoCAD, Onshape, EasyEDA, SimScale, MATLAB, LTSpice
Languages: English (L1 native), Mandarin (C1), French (B2)

Sean Lin

Contact Information

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Academic Interests

Spacecraft Guidance, Navigation, and Control; ADCS; Space-Situational-Awareness; Vehicle Autonomy; Control Theory

Education

Stanford University, Stanford, CA

M.S. Aeronautics and Astronautics, Expected December 2026

- GPA: 3.86/4.00

University of California Berkeley, Berkeley, CA

B.S. Mechanical Engineering, May 2023
B.S. Business Administration (Haas), May 2023
Minor Electrical Engineering, May 2023

- High Honors
- GPA: 3.93/4.00

Research Experience

iFR @ University of Stuttgart
P.I. Cunis Torbjørn — Jun 2026 - Present

- Controllability-based design of VLEO satellites in rarified atmospheric orbit.
- Designed an optimization workflow for connecting control-surface configuration with desired vehicle torque.
- Defined key torque gradients to implement in the controller.

Space Rendez-Vous Lab @ Stanford
P.I. Simone D'Amico — Jan 2026 - Present

- Design a vision-based satellite navigation workflow for improved space-situational-awareness (SSA).
- Propagate orbits of a catalog of 20,000+ satellites and perform a detectability analysis on visible satellites.
- Designed a catalog-refining feedback workflow where observer and target satellites reinforce each-others positions.

FLOW Lab @ Berkeley
P.I. Simo Makiharju — Sept 2021 - May 2022

- Design and assembly of Magnetohydrodynamic Pump as a demonstration of Lorentz-Force pressure gradients.

The Kulik Group @ MIT
P.I. Heather Kulik — June 2018 - Sept 2018

- Visiting student computational chemistry researcher.

Teaching Experience

ME 106: Fluid Mechanics
Course Staff — Aug 2022 - Dec 2022

- Design and grade PSET and exam rubrics
- Lead weekly online discussion and section

EECS 16B: Designing Information Devices and Systems II
Course Staff — Aug 2021 - May 2022
- Design and grade PSET and exam rubrics

Industry Experience

HyLight
Propulsion and Dynamics Intern — Sep 2024 - Mar 2025
Le Plessis-Pate, FR
- Designed an electric propulsion system test bench with Torque, Thrust, Power, and Vibration readings.
- CFD aerodynamic analysis.
- Designed a buoyancy-regulator to replace off-the-shelf seals and valves.

De-Ice
Thermal Engineering Intern — Jun 2024 - Sep 2024
Boston, MA
- Designed a testing apparatus to mimic in-flight thermal conditions for full-power UHF test monitoring.
- Perform shock and vibration analysis on avionics and propose solutions to low order resonant coupling.

NASA Ames Research Center
Mechanical Design Intern — Sep 2024 - Mar 2025
Mountain View, CA
- Design and integrate legacy shock tube parts for Electric Arc Shock Tube.
- Perform high-pressure structural analysis on critical parts.

SpaceX
Associate Avionics Engineer — Jun 2023 - Sep 2023
Hawthorne, CA
- Design a Destructive Physical Analysis lab routine on avionics EEE parts. Findings lead to a faulty part containment of $45,000.
- Perform electrical Failure Analysis on flight pressure transducers and capacitors.
- Validated Raptor engine oscillators under helium environments.

Launch Engineering Intern — May 2022 - Aug 2022
Cape Canaveral, FL
- Owns 4 assemblies for Ground-Station-Equipment and launch pad hardware.
- Closed open supply chain issues for all sleeve bearings used on the launch pad.

FormFactor Inc.
Engineering Intern — May 2021 - Aug 2021
Livermore, CA
- Silicon wafer metrology and error analysis.

Hermes Robotics
Engineering Intern — May 2020 - Aug 2020
Berkeley, CA
- Mechanical design of an actuator system that autonomously shift’s a gearstick.

Grants & Awards

University of California, Berkeley
Regents and Chancellors Scholarship — Aug 2019 - May 2023
California Alumni Scholarship — Aug 2019 - May 2023

Publications

HJ. Kulik et. al., *Designing in the face of uncertainty: exploiting electronic structure and machine learning models for discovery in inorganic chemistry*, Inorganic Chemistry, Cambridge, MA, 2019.

HJ. Kulik et. al., *Seeing is believing: Experimental spin states from machine learning model structure predictions*, The Journal of Physical Chemistry, Cambridge, MA, 2020.

Languages

English (Native), Mandarin Chinese (C1), French (B2), German (A1)

Programming Languages

Python, C++, Matlab, Julia, Java, bash, LaTeX, ROS2,

References

D'Amico, Simone, Professor, Stanford University, `damicos@stanford.edu`
Cunis, Torbjorn, Lecturer, iFR Stuttgart, `torbjoern.cunis@ifr.uni-stuttgart.de`
Jaime, Herzberger, EEE Manager, SpaceX, `Jaemi.Herzberger@spacex.com`
Makiharju, Simo, Professor, UC Berkeley, `makiharju@berkeley.edu`
Trejo, Layla, Cape Launch Hardware, SpaceX, `Layla.Trejo@spacex.com`